PHARMACOECONOMICS





Lecture 3

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PHARMACOECONOMIC METHODS



Economic

Cost consequence
Cost benefit
Cost effectiveness
Cost minimization
Cost utility

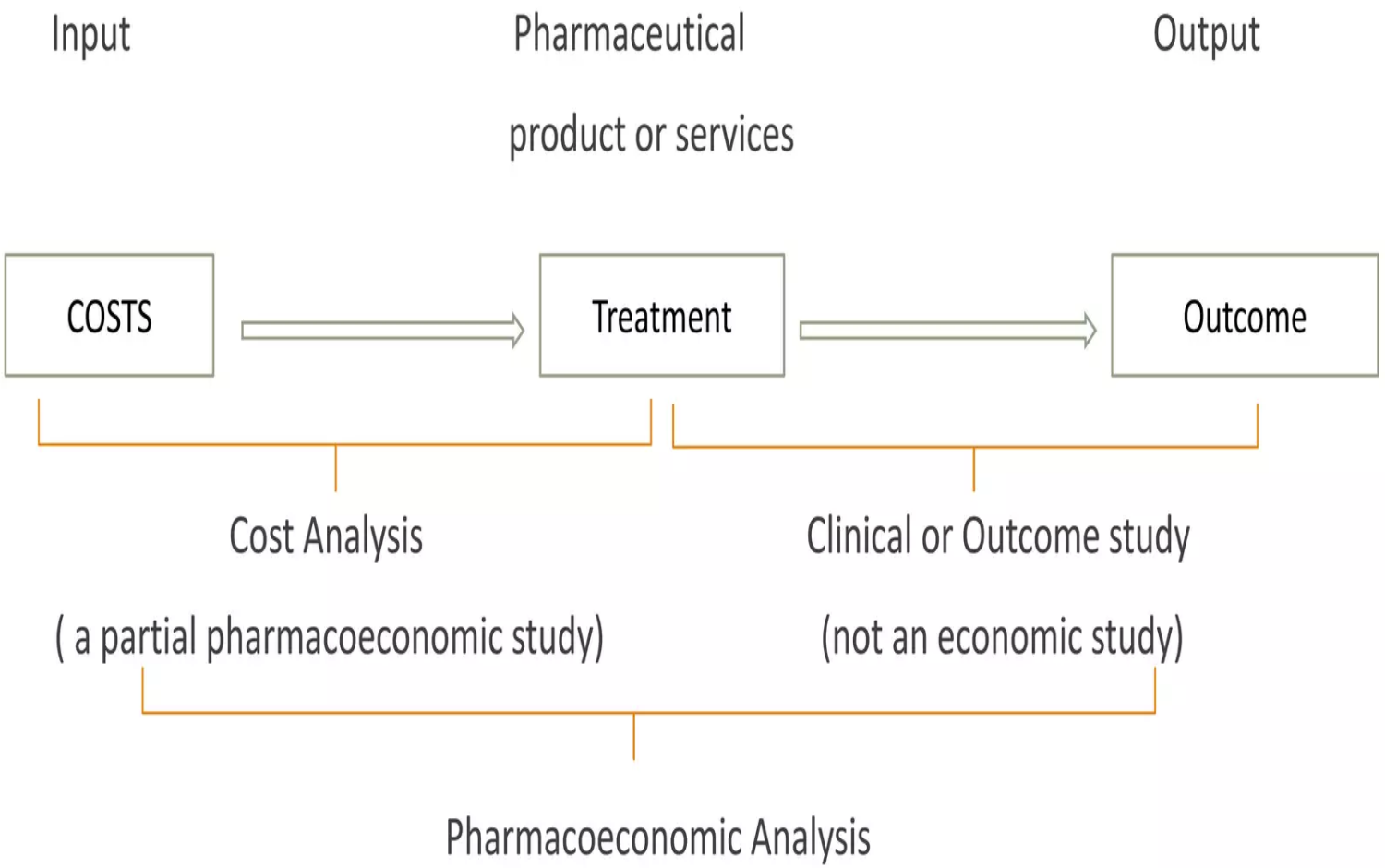
Humanistic

Quality of life
Patient preferences
Patient satisfaction

Pharmacoeconomic Methodologies

- **Economic evaluations**
  - *Partial economic evaluations*
    - Cost consequence analysis (CCA) or Cost outcome analysis (COA)
    - Cost-of-illness (COI) evaluation
  - *Full economic evaluations*
    - Cost Minimization Analysis (CMA)
    - Cost Benefit Analysis (CBA)
    - Cost Effectiveness Analysis (CEA)
    - Cost Utility Analysis (CUA)
- **Humanistic evaluation**
  - Health -Related Quality of Life (HRQOL)
  - Patient preferences
  - Patient satisfaction

Pharmacoeconomic analysis



Cost-Consequence Analysis (CCA)/(COA) :

- Is a partial economic evaluations can:

- Include simple descriptive tabulations of outcomes or resources consumed

- Costs and different types of outcomes are listed

- Require a minimum of time and effort

- A cost-outcome or cost-consequence analysis (CCA)

- Describes the costs and consequences of an alternative

- Does not provide a comparison with other treatment options

Example of a cost-consequence tabulation:

Cost components	Drug A		Drug B	
	Units	costs	units	costs
Direct medical care use/costs: Drug A/B Other drugs Physician visits Hospital stays Home care Other medical care (e.g. dialysis)				
Direct nonmedical care use/costs: Transportation Crutches and other equipment Paid caregiver time				
Indirect resource use/costs: Time missed from work for patient				

Example:

The table below shows the cost consequence analysis (CCA) for three drugs, A, B and C for treating stomach ulcers using outcome measures of SFD and % healed.

	Drug A	**Drug B**	**Drug C**
Cost	$600 per year	$210 per year	$530 per year
Outcomes			
GI SFDs	130	200	250
% Healed	50%	70%	80%

SFD: Symptoms free days or how many days on average patients didn't have gastrointestinal symptoms during the year.

Cost of Illness (COI) evaluation:

- COI identifies and estimates the overall cost of a particular disease for a defined population.
- COI evaluation method is also known as **burden of illness**.
- Has the advantage of defining the disease, its epidemiology, outcomes and consequences.
- It involves measuring the **direct** and **indirect** costs attributable to a specific disease such as diabetes, mental disorders, or cancer.

- direct costs -- the costs associated with providing treatment or prevention
- indirect costs -- the costs attributable to loss of productivity of patients with that disease or premature death

- Example – Costs of Type II diabetes in adults.

"An illness consumes resources and, thus, it has a cost. The **cost of an illness** is the sum of **three** broad components:

(1) medical resources used to treat the illness,

(2) the nonmedical resources associated with it, and

(3) lost productivity due to illness or disability"

- COI evaluation is **not used to compare competing treatment alternatives** but to provide an estimation of the financial burden of a disease. Thus the value of prevention and treatment strategies can be measured against this illness cost.

Cost Minimization Analysis (CMA):

- Cost-minimization analysis is the most basic technique.
- The objective is to identify **less costly alternative** when comparing two or more treatment alternatives.
- Compares all the relevant costs of two or more drugs.
- This type of evaluation compares two or more alternative treatments that produce **clinically equivalent outcomes.**
- Drugs must have **identical efficacy and safety** (i.e., the two alternatives must be equivalent therapeutically).
- Once this equivalency in outcome is confirmed, the costs can be identified, measured, and compared in monetary units (dollars).

- Formulary committees do this all the time (i.e., identify less costly alternative).

Examples:

Output: identical (not considered) ☐

Input: ☐ Drug #1 costs $300, ☐ Drug #2 costs $500,
☐ Drug #3 costs $200 plus $150 lab costs for monitoring.

which drug would you add to your formulary?

- If drugs A and B are antiulcer agents equivalent in efficacy and adverse drug reactions (ADRs), then the costs of using these drugs could be compared using CMA.

- Another example would be prescribing a generic preparation instead of the brand preparation.

- Or treating patient with same therapy in hospital vs home.

Another Example: CMA that compares outpatient and inpatient costs

- The estimated the costs associated with administering prostaglandin E2 gel intracervical to mothers on the day before labor was to be induced. They compared the costs of (1) application of the gel, followed by a 2-hour monitoring period and then sending the mother home for the night compared with (2) application of the gel followed by a 2-hour monitoring period and then sending the mother to the maternity unit overnight. Both groups received oxytocin the next day at the hospital to augment or induce labor.

Type of Costs	Costs for Outpatients Mean (n = 40) (SD)	Costs for Inpatients Mean (n = 36) (SD)	Statistical Difference
Labor costs	$575 (366)	$902 (482)	Yes; p = 0.002
Delivery costs	$471 (247)	$453 (236)	No; p = 0.754
Pharmacy costs	$150 (102)	$175 (139)	No; p = 0.384
Hospital costs	$3,835 (2,172)	$5,049 (2,060)	Yes; p = 0.015

COST-MINIMIZATION ANALYSIS (CMA): IN DRUG THERAPY

	Cost of therapies (₹)	
COSTS	**Drug A**	**Drug B**
Acquisition cost	250	350
Administration	75	0
Monitoring	75	25
Adverse effects	100	25
Subtotal	**500**	**400**
OUTCOMES		
Antibiotic effectiveness	90%	90%

- CMA is the simplest of the four types of pharmacoeconomics analyses because the focus is on measuring the **left-hand side** of the pharmacoeconomic equation.



FIGURE 1.1. **Basic pharmacoeconomic equation.** Pharmacoeconomic studies compare the costs (*left box*) associated with providing a pharmacy product or service (represented as Rx) to the outcome of the product or service.

- CMA is a relatively straightforward and simple method for comparing competing programs or treatment alternatives as long as the therapeutic equivalence of the alternatives being compared has been established.

- If no evidence exists to support therapeutic equivalence, then a more comprehensive method such as cost-effectiveness analysis should be employed.

Cost-minimisation analysis (CMA)

When to use	Compares two or more alternative treatments that produce clinically equivalent outcomes. Analysis thus limited to costs.
Application	• Different dosage forms of a specific active substance van with equivalent outcomes • Generics / Generic similar products • Different treatment periods with equivalent outcomes (3 days IV vs 5 days oral) • Treating patient w/same therapy in hospital vs home
Cost unit: Outcomes unit: Calculation	Monetary value Natural units (equivalent) (Cost of option A) - (Cost of option B)
Interpretation	Choose option with lowest cost

Cost Effectiveness Analysis (CEA):

- The most commonly employed method. It is a technique designed to assist a decision-maker in identifying a preferred choice among possible alternatives.

- Measures effectiveness (health benefit) in natural units (eg years of life saved, ulcers healed, weight gained, blood cholesterol level reduction) and the costs in money.

- CEA does not allow comparisons to be made between two totally **different areas of medicine with different outcomes**.

- It compares therapies with qualitatively similar outcomes (with different safety and efficacy) in a particular therapeutic area. For instance, in severe reflux oesophagitis, using a proton pump inhibitor compared to using H2 blockers.

- Typically the CEA is expressed in terms of **a ratio**; where the **denominator is a gain in health** from a measure (years of life saved, cases cured, drop in blood pressure) and the **numerator is the cost** associated with the health gain.

$$\frac{\text{Cost (\$)}}{\text{Therapeutic effect (Natural units)}}$$

Cost-Effectiveness Analysis

- Example:

	Total Cost/ 100 Patients	Lives Saved/ 100 Patients	Average CE Ratio
Drug A	$220,000	79	$2784.81/ life saved
Drug B	$20,000	78	$256.41/ life saved

- The results of CEA are expressed as a ratio either as:

1- an average cost-effectiveness ratio **(ACER)**

2- or as an incremental cost effectiveness ratio **(ICER).**

- **An ACER** represents the total cost of a program or treatment alternative divided by its clinical outcome to yield a ratio representing the (dollar) cost per specific clinical outcome gained independent of comparators.

Average cost effectiveness ratio (ACER) =

Health care Cost ($)/clinical outcome (in Natural units)

- Using this ratio (ACER), the clinician would choose the alternative with **the least cost per outcome gained**.

CEA IN DRUG THERAPY

	Cost of therapies (₹)	
COSTS	**Drug A**	**Drug B**
Acquisition cost	300	400
Administration	50	0
Monitoring	50	0
Adverse effects	100	0
Subtotal	500	400
OUTPUTS		
Extra years of life	**2.22**	**1.6**
Cost-effectiveness ratio	500/2.2 = ₹ 225	400/1.6 = ₹ 250
		Per extra year of life

- The most cost-effective alternative is **not always the least costly alternative.** In this regard, cost-effectiveness **need not** be **cost reduction** but rather **cost optimization** (i.e. ensure the value or outcome achieved for the amount of money spent. It's less about spending as little as possible and more about ensuring that each dollar spent contributes significantly to the desired outcome).

- Often clinical effectiveness is gained at an increased cost. **ICER** can be used to determine the additional cost and effectiveness gained when one treatment alternative is compared with the next best treatment alternative.

- Thus, instead of comparing the ACERs of each treatment alternative, the additional cost that a treatment alternative has over another treatment is compared with the additional effect or outcome it provides.

- The ICER can be summarized as follows:

- The key measure of CEA is the incremental cost effectiveness ratio (ICER).

Incremental Cost Effectiveness Ratio (ICER) =

(Cost of drug A-Cost of drug B)/(Benefits of drug A-Benefits of drug B)

ICER = Difference in costs (A-B)/Difference in benefits (A-B)

- This formula yields the **additional cost** required to obtain the **additional effect** gained by switching from drug A (current) to drug B (comparator) (ICER for strategy B compared to strategy A).

Cost-effectiveness Analysis

- The most commonly CERs used are the:
  - Average cost-effectiveness ratio (ACER)

$$ACER = \frac{CostB}{EffectivenessB}$$

  - Incremental cost-effectiveness ratio (ICER)

$$ICER = \frac{C_1 - C_2}{E_1 - E_2} = \frac{\Delta C}{\Delta E}$$

Table 1 Cost and benefit for each strategy in the reference case

Strategy	Cost	Benefit (QALYs)
A	$600,000	24
B	$1,300,000	34
C	$400,000	16
D	$500,000	26
E	$700,000	20
F	$100,000	6
G	$1,000,000	30
H	$350,000	8

Benefit is measured in quality-adjusted life-years (QALYs)

From Table 1, the cost of strategy A is $600,000 and the cost of strategy B is $1,300,000, so the incremental cost of strategy B compared to strategy A is $700,000.

Meanwhile, the benefit of strategy A is 24 QALYs and the benefit of strategy B is 34 QALYs, so the incremental benefit of strategy B compared to strategy A is 10 QALYs. Dividing the incremental cost of $700,000 by the incremental benefit of 10 QALYs results in an ICER of $70,000 per QALY for strategy B compared to strategy A.

Cost-effectiveness Example

	Cost	Effectiveness	Avg. C/E Ratio	Incremental C/E Ratio
Drug A	$50	30 mg/dl	50/30	
Drug B	$70	40 mg/dl	70/40	(70-50)/(40-30)



Cost-effectiveness Example

	Cost	Effectiveness	Avg. C/E Ratio	Incremental C/E Ratio
Drug A	$50	30 mg/dl	$1.67per mg/dl	
Drug B	$70	40 mg/dl	$1.75per mg/dl	$2 per additional mg/dl

- CEA can provide valuable data to support drug policy, formulary management, and individual patient treatment decisions. Globally, CEA is being used to set public policies regarding the use of pharmaceutical products (national formularies) in countries such as Australia, New Zealand, and Canada.

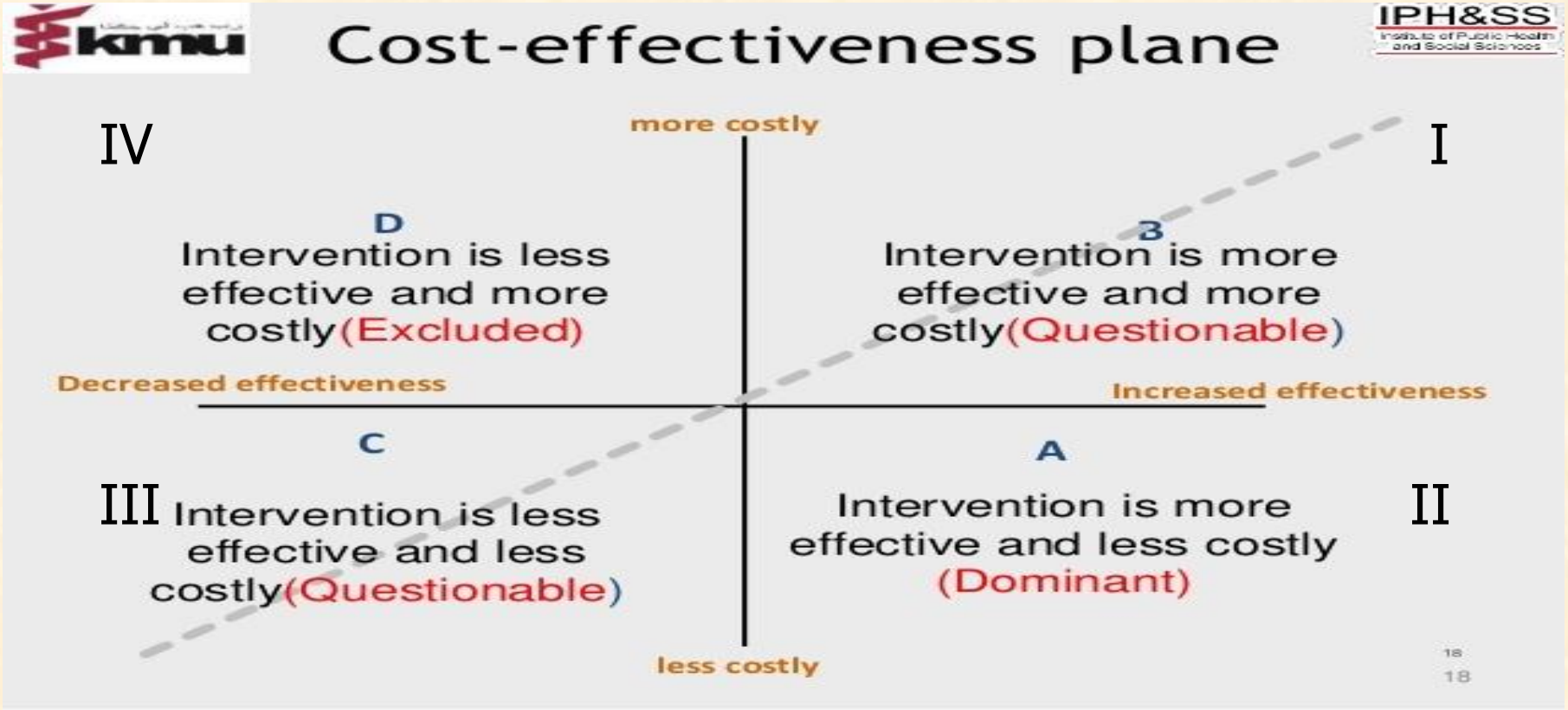
Examples where CEA is best technique

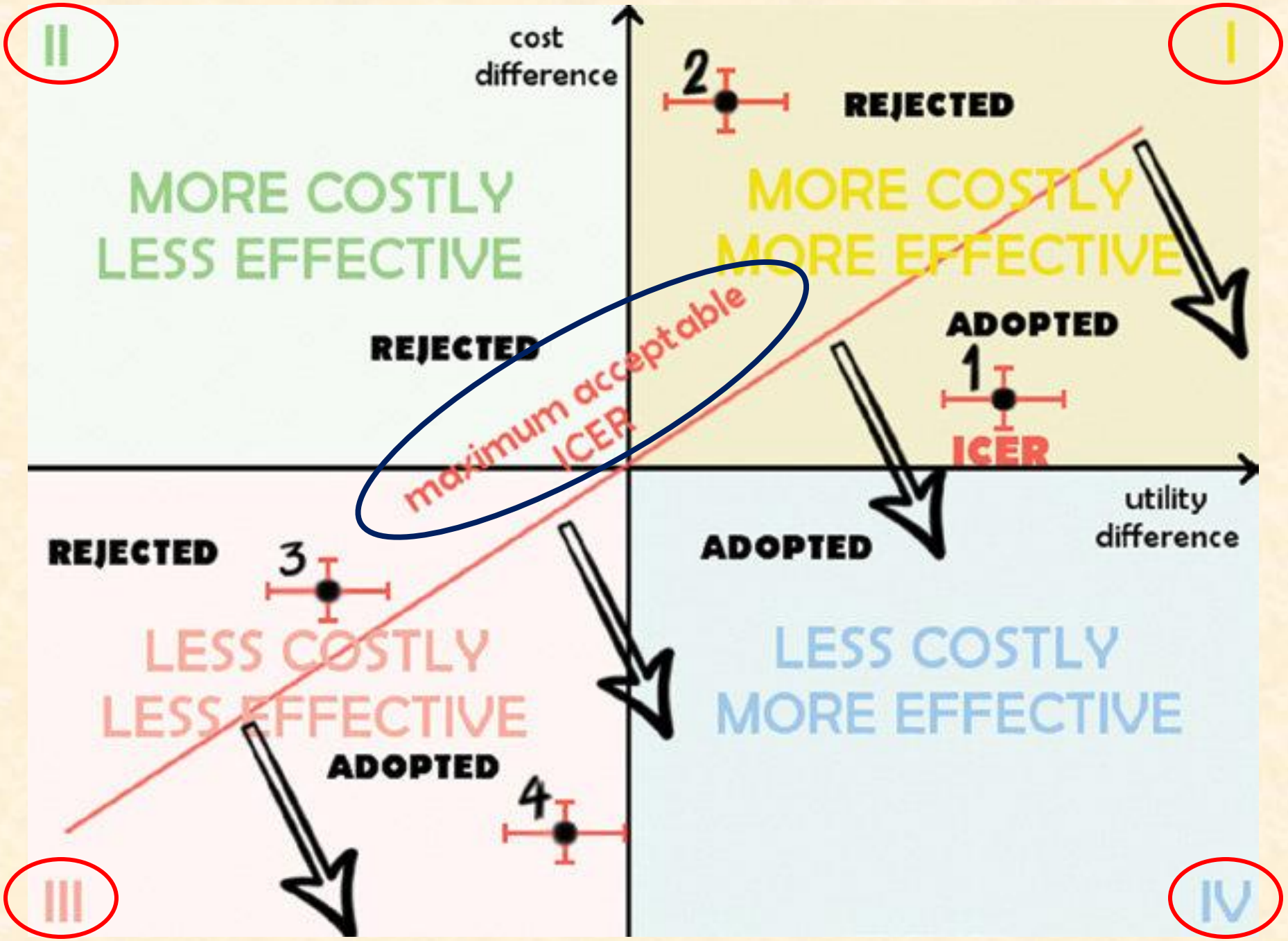
- Compare the costs and outcomes of two or more antihypertensives
  - Cost per decrease in blood pressure
- Compare two programs designed to prevent excess mortality
  - Cost per life saved

Note: **Cost-utility analysis (CUA)** is a special form of cost-effectiveness analysis

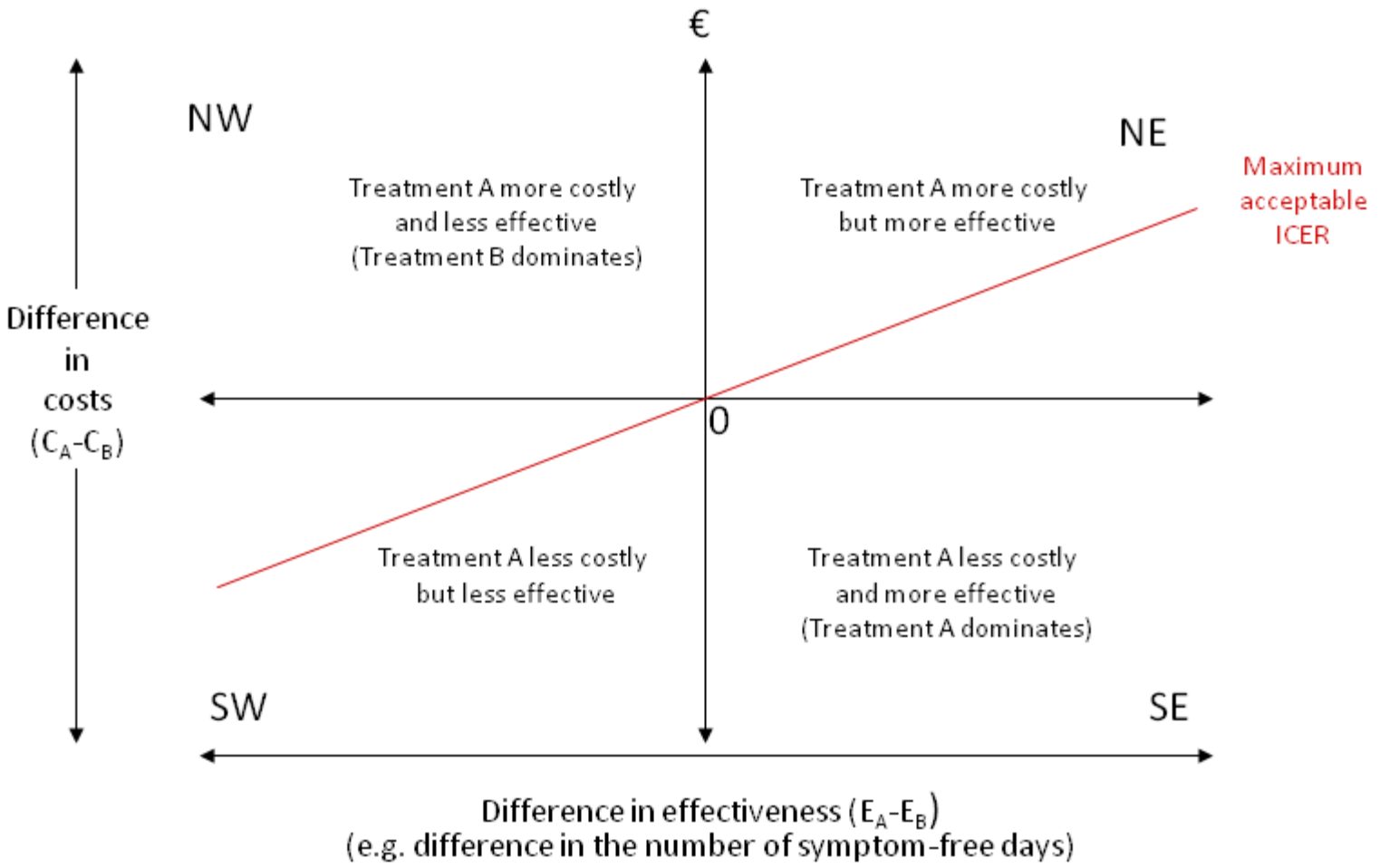
- Cost-effectiveness analyses are often visualized on a cost-effectiveness plane (scale) consisting of **four-quadrants**.

- Outcomes plotted in Quadrant I are more effective and more expensive, those in Quadrant II are more effective and less expensive, those in Quadrant III are less effective and less expensive, and those in Quadrant IV are less effective and more expensive.




II
cost difference
I
2
REJECTED
MORE COSTLY
LESS EFFECTIVE
MORE COSTLY
MORE EFFECTIVE
ADOPTED
1
ICER
REJECTED
maximum acceptable ICER
utility difference
REJECTED
3
ADOPTED
LESS COSTLY
LESS EFFECTIVE
LESS COSTLY
MORE EFFECTIVE
ADOPTED
4
III
IV

The incremental cost-effectiveness plane of treatments A and B.



COST EFFECTIVENESS GRID (network)

Cost effectiveness	Lower cost	Same cost	Higher cost
Lower effectiveness	A	B	C
Same effectiveness	D	E	F
Higher effectiveness	G	H	I

are cost effective choices

are not cost effective

- **For cell A and Cell I, ICER is calculated to determine the extra cost for each extra unit of outcome.**

Cost and Effectiveness Comparison Grid

Effectiveness				
Cost		A>B	A=B	A<B
	A>B	***Analyze***	**Choose B**	**Choose B**
	A=B	**Choose A**	**Indifferent**	**Choose B**
	A<B	**Choose A**	**Choose A**	***Analyze***

The cost-effectiveness grid:

- can be used to illustrate the definition of "cost□effectiveness".

- To determine whether a therapy or service is cost-effective, both the costs and the effectiveness must be considered. If the new treatment (1) is both more effective and less costly (cell G), (2) is more effective at the same price (cell H), or (3) has the same effectiveness at a lower price (cell D), then it is considered **cost-effective** (blue shaded cells).

- On the other hand, if the new drug (1) is less effective and more costly (cell C), (2) has the same effectiveness but costs more (cell F), or (3) has lower effectiveness for the same costs (cell B), then it is **not cost-effective** (yellow shaded cells).

- There are three other possibilities (cells with no shading); the new drug (1) is more expensive and more effective (cell I) (**a very common finding**), (2) is less expensive but less effective (cell A), or (3) has the same price and the same effectiveness as the standard product (cell E).

- For the middle cell E, other factors may be considered to determine which medication might be best.

- For the other two cells, ICER is calculated to determine the extra cost for each extra unit of outcome.

- It is left up to the readers to determine whether they think the new product is "cost-effective" on the basis of value judgment. **The underlying subjectivity** as to whether the added benefit is worth the added cost is a disadvantage of CEA.


THANK
YOU